A Detailed Characterization of Starlink One-way Delay

Johan Garcia
Karlstad University
Karlstad, Sweden
johan.garcia@kau.se

Simon Sundberg
Karlstad University
Karlstad, Sweden
simon.sundberg@kau.se

Anna Brunstrom
Karlstad University
Karlstad, Sweden
anna.brunstrom@kau.se

Abstract

Low Earth Orbit (LEO) satellite networks, such as Starlink, are transforming global Internet access by delivering high-speed connectivity to underserved and remote regions. Despite extensive research into Starlink's performance, latency characteristics remain underexplored. This study presents a comprehensive analysis of one-way delay components in the Starlink network using high-frequency, high-precision measurement probes. Over a 10-day period, more than 500 million probe packets were collected and analyzed. The results reveal minor diurnal latency variation and provide means to separate out the delay components contributing to the observed one-way delay, and we sketch a delay model and provide empirical distributions. By measuring both uplink and downlink paths, the study uncovers significant differences in scheduling behavior, with uplink delays more affected by Starlink's periodic 15-second reconfiguration cycles. The results also highlight the limitations of using too coarse measurement intervals, which can introduce aliasing effects. Our OWD data set and traffic generation tool are made available to support further research in the area.

CCS Concepts

• **Networks** → **Network measurement**; **Network performance modeling**; *Network performance analysis*; Wireless access networks.

Keywords

Starlink, Delay modeling, Network Measurements, Satellite Networks, Performance Evaluation

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1 Introduction

Low Earth Orbit (LEO) satellite communication systems have the ability to provide high-speed Internet connectivity across the globe, serving remote and sparsely populated areas that previously faced considerable challenges in terms of Internet access. The primary LEO satellite operator is Starlink, which currently has more than 5 million customers [39]. Starlink has been the subject of a number of studies aimed at characterizing performance aspects in general, and in relation to specifics such as the effects of weather [4, 25, 28], the impact of mobility [3, 13, 17, 23, 24, 42], the interaction with TCP congestion control [7, 8, 11, 22] and the use of multi-connectivity [3, 17, 26, 27]. So far, there has been no detailed and fine-grained study of delay aspects. In this study, we utilize high-frequency, high-precision measurement probes to perform a detailed characterization of the one-way delay components in the Starlink network.

We perform a 10-day measurement campaign encompassing more than half a billion probe packets. Analyzing these measurements, we find that there are indications of only minor amounts of diurnal variations in latency. We further provide a sketch of a delay model incorporating several Starlink-specific delay components. As we simultaneously measure both the uplink and downlink, we can compare their latency characteristics. Our measurements show that the scheduling behavior differs considerably between uplink and downlink, and that the delay impacts of the periodic 15-second reconfigurations are larger for the uplink than the downlink.

We provide the following contributions:

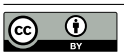
- a characterization of Starlink one-way delay across multiple time-scales,
- a delay model detailing the components of observed delay variations and empirical distributions related to three major delay components,
- an analysis showing that using a 10 or 20 ms measurement interval may result in misleading aliasing effects.

We believe that the insights on delay characteristics provided by this work can be helpful when designing simulation or emulation tools that aim to reproduce the behavior of Starlink or other LEO systems.

2 Background and Related Work

Previous research has established that the Starlink network characteristics vary over time in terms of available capacity, round-trip time, and packet loss [12, 22, 30]. Starlink uses a 15 second reconfiguration interval [12, 30] and schedules transmissions in $1.\overline{33}$ ms link layer frames [10, 18]. On short timescales the throughput of Starlink is dependent on the physical transmission rate, and the allocation structure of the link layer frames [9].

Many studies report Starlink delay results by characterizing the distribution of observed RTT by means of boxplots, ECDFs, or evolution over time [3, 5, 15, 26, 33, 34, 38, 45, 47]. Early RTT results were reported in [29], and a large-scale examination of Starlink latency with more than 2400 vantage points was performed in [19] to examine aspects such as inter-satellite-link (ISL) delay impact. The impact of the global 15-second reconfiguration events on measured RTT is examined in [22], and they also examine the RTT impact from local reconfigurations where satellites are switched within a 15 s period. In relation to previous work that considers delay or latency, our study differs mainly in terms of the measurement



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resolution and the number of collected observations, which allows for better characterization and modeling of the delay components.

Some studies have used simulation or emulation to evaluate Starlink performance [2, 16, 20–22, 41]. Since modeling the delays is in many cases important for simulation and emulation fidelity, the results presented here can be beneficial when setting up future simulation or emulation studies. that aim to faithfully represent Starlink connectivity. In the context of 5G, other work has examined approaches for delay measurements [14, 31], or characterized the delay in 5G networks [1, 36].

3 Measurement collection

For our measurements, we use a Gen-2 Starlink user terminal (UT) connected via Ethernet to our measurement computer which is equipped with an Intel X550-T2 Network Interface Card (NIC). This NIC has two Ethernet ports and is capable of providing hardware timestamps for both received and transmitted packets. To generate the traffic, we utilize a custom traffic generator, nanoprobe, which is loosely based on the existing nanoping tool. Nanoprobe has been designed to allow high-frequency generation of probe packets while also extracting the hardware timestamps for both received and transmitted packets[1]. We set up the measurements so that one port of the NIC is connected to the Starlink terminal, and the other port has a public IP address.

Nanoprobe is configured in a client-server setup. The nanoprobe client is attached to the port connected to the Starlink UT, and the nanoprobe server is attached to the port connected to the Internet. We configure nanoprobe to simultaneously, and independently, send probe packets from server to client (downlink, DL) and from client to server (uplink, UL). Probe packets are sent with an inter-packet time of 266 μs, which corresponds to a packet rate of 3750 packets per second. As the packets are 150 bytes including the UDP and IP headers, this corresponds to a bit rate of 4.5 Mbps.

The objective here is to characterize latency aspects under a lightly loaded scenario, where the offered measurement load is well below Starlink capacity. Typical applications generating such latency-sensitive and non-greedy traffic would be video-conferencing, gaming, etc. Greedy and capacity-seeking traffic, as is the case for long-lived TCP connections, would result in higher rates, and additional queuing delay. A number of studies report Starlink downlink throughput in the range of 50 to 250 Mbps, and thus the offered rate for our latency measurements is well below what the system is capable of providing. For the uplink, there are fewer studies, but here too the observed empirical throughput is significantly above our measurement traffic rate.

The measurements are collected in separate runs, where each run is 50 seconds long, and a new run is performed every 10 minutes during the measurement period. The 50-second run duration enables the study of effects at the 15-second reconfiguration interval. Over a 10-day period in April 2025 we collected 1440 measurement runs, of which 3 could not be successfully completed due to spurious issues with the nanoprobe initial handshake negotiation[2].

With our measurement setup, one-way delay measurements can be performed with very high precision. Since both the transmit and receive timestamps are hardware-generated and derived from the same on-NIC 80 MHz oscillator, there is no relative clock skew between sending and receiving timestamps, and the timing measurements have a precision in the order of tens of nanoseconds. While our measurement setup includes the traversal of the regular Internet from our server to the Starlink Point of Presence (PoP) in Frankfurt, this connection is well-provisioned and does not significantly hinder our ability to detect Starlink-specific timing patterns with sub-millisecond precision, as illustrated later.

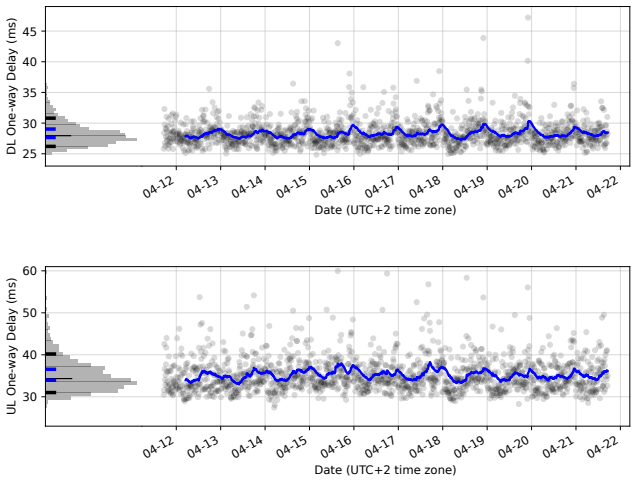


Figure 1: Per-run mean OWD for downlink (DL, upper) and uplink (UL, lower).

4 Day-scale variation

We initially consider the delay characteristics on larger timescales. Figure 1 provides a plot of the mean one-way delay for all the runs during the measurement campaign. Included is an exponentially weighted moving average (EWMA) with a halflife of 3 hours that makes visual identification of diurnal trends easier. As can be seen in the figure, the vast majority of runs have a mean downlink one-way delay (OWD) between 25 and 35 ms, while for uplink, the range is broader, lying between 30 and 50 ms. These values include a baseline delay, d_{Base}, for the Internet traversal from our server to the Starlink PoP, which we find to be around 11 ms. In the distribution to the left, the thick marks show the 5th and 95th percentiles of the OWD (black) and the EWMA values (blue), while the tall thin line indicates the OWD median.

In terms of diurnal variation, the EWMA shows some fluctuations, with the lowest delays consistently occurring during the early morning hours. Decreasing RTTs during morning hours is reported in [43], while other studies report small [35] or no [25] diurnal variation, although the details of the measurement setup differ between studies. The delay variation observed here could possibly result from the network connection between the measurement server and the Starlink PoP, but this is considered unlikely. If the diurnal effects were due to the PoP to server segment, then it would be reasonable to expect that the magnitude of the one-way delay variation would be similar in both directions. However, as

[1]Nanoprobe is available at https://github.com/simosund/nanoprobe.
[2]The OWD data is available at https://doi.org/10.5281/zenodo.16275284

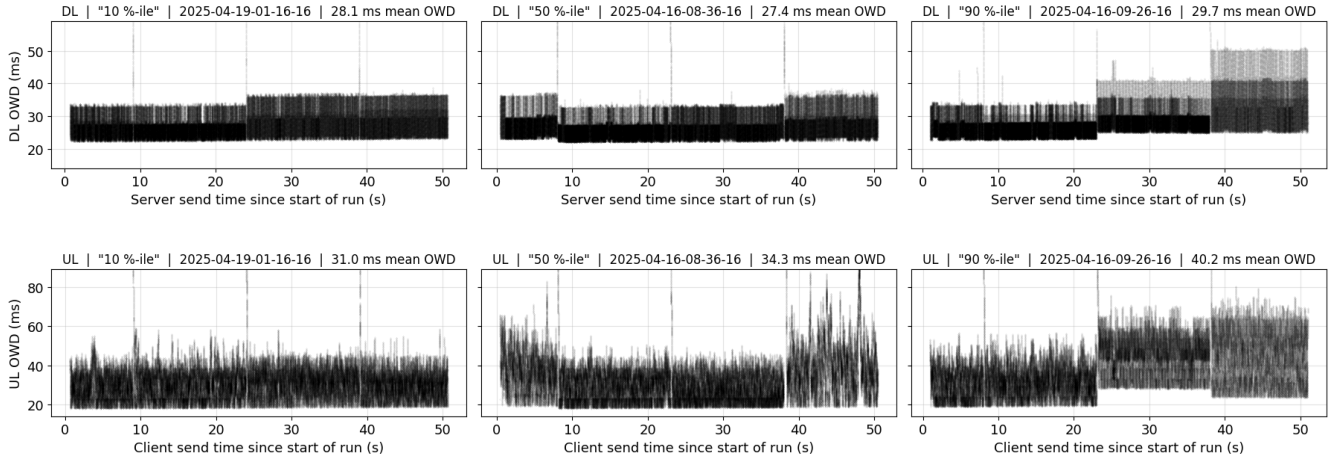


Figure 2: Within-run OWD sample values from 3 selected runs (10, 50, 90th percentile of per-run mean DL/UL OWD). All of the approximately 187K samples per subfigure are plotted.

can be seen in the graph, the EWMA delay difference between the percentiles is larger for uplink than for downlink. This suggests that the diurnal variation is tied to the Starlink segment, given that the difference in distribution width between uplink and downlink follows the same relative magnitude as the EWMA difference.

5 Second-scale variation

We now move to a smaller timescale and consider the behavior within individual measurement runs. The OWD of the approximately 187,000 probe packets in a run are shown in Figure 2 for three selected runs. These selected runs correspond to those with a mean UL delay at the 10th, 50th, and 90th percentiles of all measurement runs.

There are several notable aspects in these graphs. A periodic 15-second reconfiguration event is visible in all subgraphs, with distinct vertical lines occurring three times in each subgraph. Note that the x-axis time is relative to the start time of the run, not seconds in the minute. Previous research [3, 22, 40] has found that reconfigurations happen at the well-known times 12, 27, 42, and 57 seconds in the minute.

In addition to the regular increases in OWD occurring at the reconfiguration events, several of the subgraphs also show that delay characteristics change in the segments on either side of the reconfiguration event. This is expected, given that during reconfiguration, the allocation of communication resources and satellites often changes. It can be noted that two distinct types of delay changes can be seen in the subgraphs. There is some variation in the minimum delay observed within each segment, which varies by up to 10 ms in these particular subfigures, with the most variation seen in the uplink 90th percentile case.

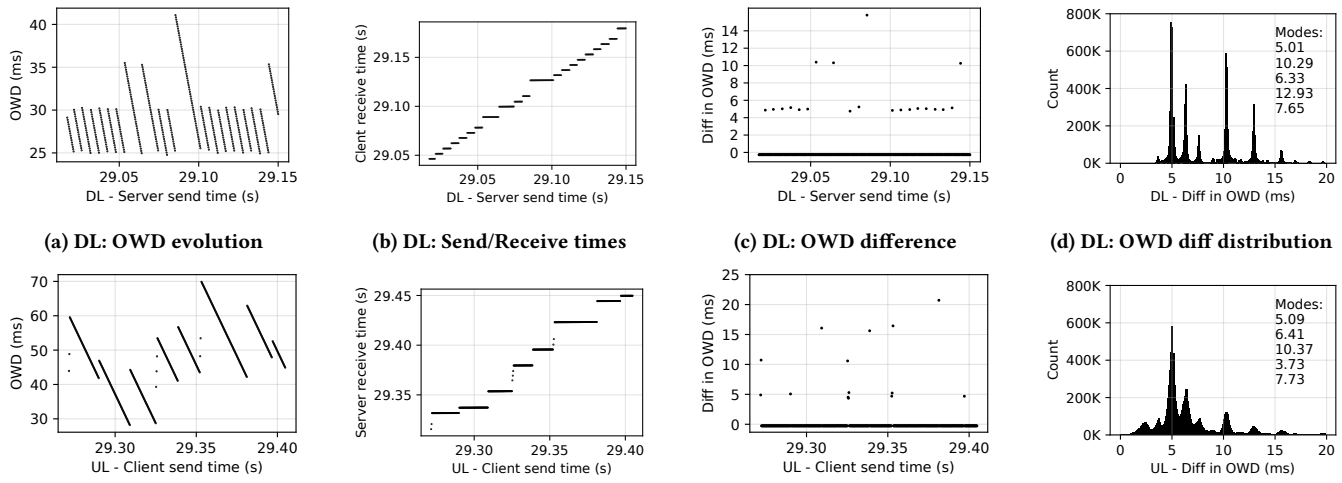
Another structural delay aspect is the range over which the delays vary within a segment. The amount of variation ranges from approximately 10 ms in the downlink 10th percentile subfigure to up to 40 ms for the uplink 90th percentile subfigure. There are also changes in the overall amount of variability present. The downlink measurements exhibit a clear structural pattern with small amounts of variation, while the uplink measurements show larger amounts of variation. There are, however, indications of structure visible also in the uplink measurements, and as will be shown, there are considerable structural components present in the uplink delay variation as well.

6 Millisecond-scale variation

To further examine the structure observed within the measurement runs, we consider the OWD patterns at even smaller timescales in Figure 3. Figures 3a and 3e show a small segment of observations from one of the runs presented in Figure 2. With the x-axis now stretched and displaying only 150 ms, a clearer pattern emerges. It can be seen that there are regular streaks where the OWD decreases for the subsequently sent packets.

Recall that the probe packets were sent very frequently, with one packet every 266 microseconds. To understand the reason for this downward-sloping OWD evolution, Figures 3b and 3f provide an alternate representation of the same data. On the x-axis, we again show the send time of the probe packets, but the y-axis now shows the receive time. It becomes evident that large chunks of data are received at a small number of distinct receive times. This is consistent with earlier research, which has established that Starlink uses a $1.\overline{33}$ ms link layer slot time. The spacing between received data points aligns with this interval, and this pattern is decidedly a result of Starlink's link layer scheduling.

Coming back to the data shown in Figures 3a and 3e, one can note that another interesting way to analyze the data is to consider the difference in OWD between subsequent probe packets. In a downward slope, the OWD is decreasing, so the OWD difference will be negative. Figures 3c and 3g show this OWD difference, and the line just below 0 represents all the probe packets that are part of a downward streak. The mean value of this OWD difference is -264 μs, which corresponds to the sending interval and the burst rate at which probe packets can be delivered during the horizontal sections of Figures 3b and 3f.



(a) DL: OWD evolution (b) DL: Send/Receive times (c) DL: OWD difference (d) DL: OWD diff distribution

(e) UL: OWD evolution (f) UL: Send/Receive times (g) UL: OWD difference (h) UL: OWD diff distribution

Figure 3: 150 ms zoom-in of run 2025-04-16-09-26-16 (a-c, e-g). Distribution of OWD diff (d, h).

There is also a small subset of packets that exhibit a higher difference in OWD. These are the first packets to be received within a Starlink transmission burst, and they are the ones that experience the largest scheduling waiting time, d_{sched}. So, while the vast majority of packets have a difference value of around −264 μs, one packet per transmission burst will have a larger difference value that corresponds to the highest d_{sched} for that burst.

We now process the complete dataset in order to compute the OWD difference value for each of the half-billion packets, and we show the distribution of the difference values larger than zero in Figures 3d and 3h. Also shown are the OWD difference values corresponding to the five highest peaks in the distribution, derived as the top five modes with a 0.01 ms bin resolution. There is a noteworthy consistency in the placement of the peaks, where all inter-peak distances are very close to integer multiples of $1.\overline{33}$ ms, reflecting the Starlink link layer frame time. Thus, it is clear that our measurement setup is able to resolve these Starlink-specific details also on the uplink, where the Internet traversal comes after the Starlink traversal.

Comparing the behavior for the uplink, as shown in Figures 3e-3h, to the downlink in Figures 3a-3d, there are both similarities and differences. The overall structure is similar, but there is more variation in the uplink scheduling which has been attributed to the assumed poll-randomize-grant operation of the uplink [34]. The slopes vary more in terms of their length and minimum value. However, the slopes are well-defined straight lines, indicating that there is no significant variation introduced by the fixed network connection from the Starlink PoP to the measurement computer during a send burst.

While the larger variations in the streaks are reflected in Figures 3f and 3g, when one considers the distribution across the whole dataset—as shown in Figure 3h—it is noticeable that, also for the uplink, the Starlink-specific timing of $1.\overline{33}$ ms is clearly reflected in the distribution. This again indicates that the fixed network does not contribute large amounts of measurement noise.

7 Reconfiguration delay anomaly

An important aspect that was hinted at in the data shown in Figure 2 is the increase in one-way delay during the reconfiguration of transmission resources, which occurs every 15 seconds. Determining the duration and delay characteristics of these reallocation events can be useful, for example in conjunction with congestion control schemes that explicitly compensate for such anomalies [44]. Figure 5 provides a zoom-in of the data from Figure 2 for the periods where reconfiguration occurs, with the y-axis now scaled to show the highest observed OWD values.

It is clear that reconfiguration events have a considerable impact on the one-way delay, with the maximum observed one-way delay reaching around 90 ms for the downlink and up to 200 ms for the uplink in these examples. It can also be seen that the duration during which the OWD is impacted by the reconfiguration varies, ranging approximately from 100 ms up to 400 ms.

To provide a wider characterization of the OWD inflation during the reconfiguration phase, we extract the maximum OWD for each reconfiguration event in our measurements. For our processing we bound a reconfiguration event by conservative timing boundaries of ± 1.5 s around the well known reconfiguration times. Figure 4 shows the distribution of the highest observed OWD for the 4311 reconfiguration events in our data. We note there are also losses present during reconfiguration, but we leave a detailed characterization of Starlink losses to future work.

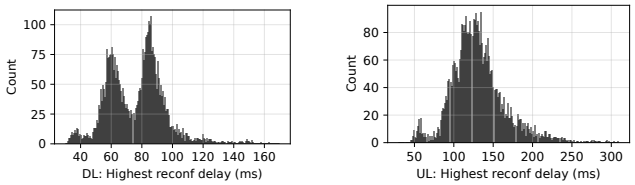


Figure 4: Max OWD within the reconfiguration event

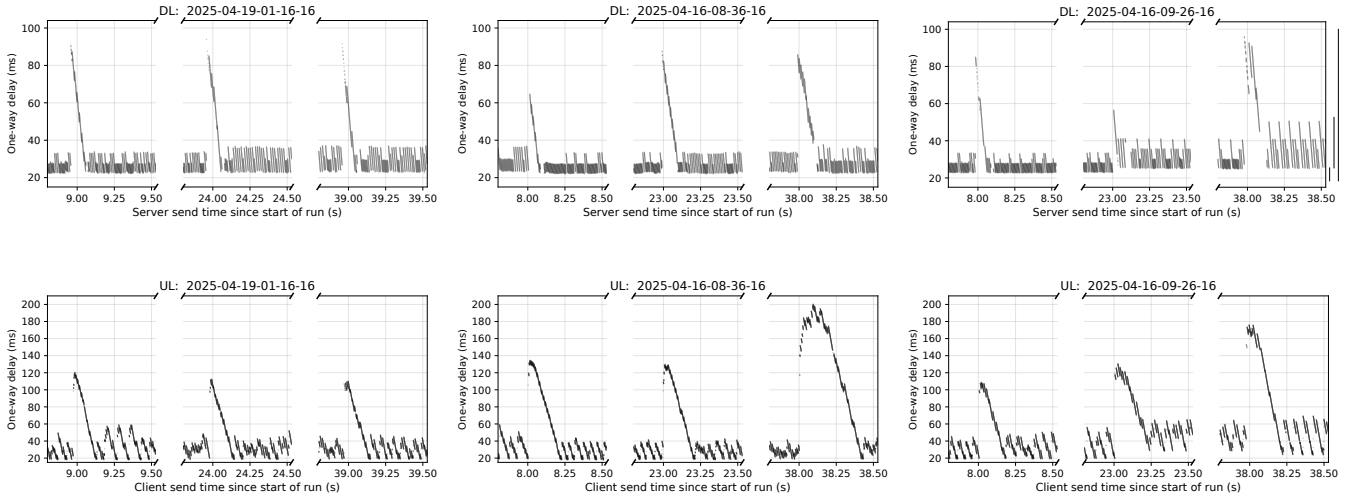


Figure 5: OWD evolution during the reconfiguration events occuring at 15 s intervals

8 Delay variation modelling

Given these insights into the structure and characteristics of the Starlink one-way delay, we can now sketch a high-level model that decomposes the observed delay into components as follows: $d_{\text{OWD}} = d_{\text{Base}} + d_{\text{BaseVar}} + d_{\text{SLfix}} + (1 - R) \cdot (d_{\text{SLsatVar}} + d_{\text{sched}}) + R \cdot d_{\text{reconfQ}}$. Of these delay components, d_{Base} is the minimal, constant, baseline delay from the server to the Starlink PoP, and d_{BaseVar} represents all sources of dynamic variation between server and PoP.

The Starlink internal fixed delay d_{SLfix}, corresponds to the strict minimum of time required to send a packet from the Starlink PoP to the ground station and up through the satellite bent-pipe to the Starlink user terminal and client. The d_{SLfix} time represents idealized circumstances, where the serving satellite is delay-optimally positioned, the serving ground station is appropriately selected, and processing delays are minimal. In practice, the satellites are constantly moving and not optimally placed, and traffic may also be reallocated between different ground stations, or inter-satellite links may be employed. The delay variation caused by these sources are captured in d_{SLsatVar}, which captures the same variability as the minRTT concept discussed in [5].

The effect of this d_{SLsatVar} component, is visible in Figure 2 and in particular in the uplink 90th percentile subplot where the magnitude of d_{SLsatVar} is illustrated. As we cannot directly observe d_{SLfix}, it is here approximated by the minimum observed value across the quarter billion observed OWD values in each direction. With this approximation, we can now compute a per segment value $d^j_{SLsatVar}$ as $d^j_{SLsatVar} = min(d_{x,j}) - d_{\text{SLfix}}$ where x is the index of the approximately 45000 observations within the 12 s non-reconfiguration parts of segment j.

The resulting d_{SLsatVar} distribution is shown in Figure 6. We can note that this d_{SLsatVar} delay component is considerably smaller than the d_{sched} component, i.e. the time a packet has to wait before being scheduled in a link frame. The underlying reason for the multiple peaks observed in the d_{SLsatVar} distribution is currently undetermined. One speculative explanation is that the traffic may be routed to different ground stations, but there may also be other underlying reasons [6]. Finally, the d_{reconfQ} represents the reconfiguration specific delay that results from additional queuing during reconfiguration, as characterized in the previous section. The R variable is a boolean that indicates whether the link is in the reconfiguration state or the normal state. A visual illustration of d_{SLsatVar}, d_{sched}, and d_{reconfQ} are provided by the three lines to the right of the DL rightmost subfigure in Figure 5. We note that our model can be helpful when reasoning about the one-way delay components as is, but it could benefit from further elaboration. We leave this for future work.

An important aspect of measuring and modeling is the sampling rate, i.e. the frequency of observations. As Nyquist and Shannon have shown [32, 37], when studying a continuous periodic signal, the sampling rate needs to be at least twice the highest frequency present in the signal in order to resolve its structure. In studies of Starlink delay the signal is not continuous, but strong periodicity is present related to the $1.\overline{33}$ ms link layer frame time. In order to fully resolve all details, measurements should therefore be performed more frequently than $1.\overline{33}$ ms. Measuring periodic signals with a rate below the Nyquist rate creates the risk of aliasing, which may lead to an incomplete or misleading understanding of the studied phenomenon.

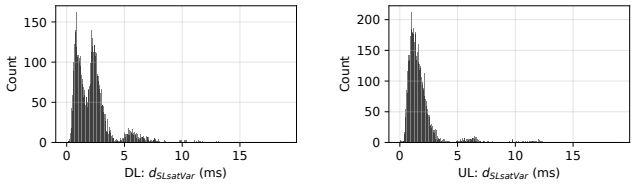


Figure 6: Starlink variable delay component

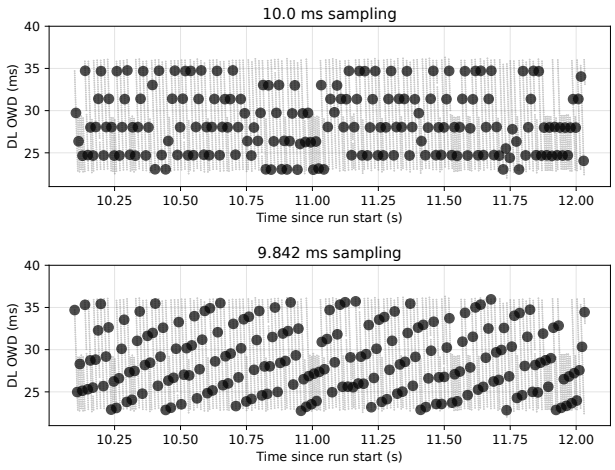


Figure 7: Two examples of aliasing effects for the same data. Simultaneous measurements at both 266 μs intervals (small dots), and at 10 ms and 9.842 ms intervals (large dots)

We provide two visual examples of the aliasing that can occur when measuring one-way delay at around 10 ms measurement intervals in Figure 7. For the upper subfigure, we use two concurrent nanoprobe instances to independently collect OWD samples with 266 μs and 10 ms sampling intervals. The figure provides a visual example of the horizontal banding that can occur when 10 ms sampling is used. One prior study using 20 ms sampling has exhibited such aliasing-induced banding [40] and two studies measuring at 10 ms intervals [34, 46] report some DL OWD banding. Similar aliasing-induced banding is implied for 5G in [14]. However, these effects are aliasing artifacts and do not reflect true underlying behavior. A slight variation in measurement interval could get a very different aliasing patterns.

In the lower subfigure, we illustrate the effect of a change in sampling rate. For this figure the underlying data is the same as above, but for the coarse data a sampling interval of 9.842 ms is now constructed by 37-fold subsampling of the 266 μs observations. This sample rate, which is only 1.6% lower than the 10¨ms employed in the upper subfigure, now results in a markedly different visual appearance with overlapping upwards sloping streaks.

9 Conclusions

In this work we analyze over 500 million high-precision probe packets to provide a fine-grained characterization of one-way delay in the Starlink network. We identify and examine three delay sources that contribute to the delay: sat-path variability, scheduling wait, and reconfiguration queuing. Uplink and downlink paths exhibit distinct scheduling behaviors, with uplink delays more affected by Starlink's 15-second reconfiguration cycles. The study also reveals that coarse measurement intervals, as often employed in previous works, can distort fine-grained structural latency analysis.

Our findings offer an increased understanding of the operations of Starlink as well as valuable insights for the development of accurate simulation and emulation tools for Starlink and similar LEO systems.

Acknowledgments

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